

JATIN CHOPRA

Personal Information

Full Name: Jatin Chopra **Nationality:** Indian **Degree Programme:** MSc Astrophysics
Current Institution: Rheinische Friedrich-Wilhelms-Universität Bonn
(Expected) Graduation Date: 15th November 2026
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Education

Rheinische Friedrich-Wilhelms-Universität Bonn Bonn, Germany
MSc Astrophysics October 2023 – November 2026

Savitribai Phule Pune University Pune, India
BSc Physics (Minor in Mathematics and Electronics) July 2019 – September 2022
CGPA: 9.77

Research Internships

- Estimating Initial Cloud Conditions from Molecular Abundances using Neural Networks and Bayesian Inference** *October 2025 – Ongoing*
 - * Supervised by Mr. Gijs Vermariën and Prof. Serena Viti at the Argelander-Institut für Astronomie (AIfA) and Universiteit Leiden. Developing a Neural Network in **JAX** and using **NumPyro** to predict molecular abundances in interstellar gas clouds.
 - * Performing Bayesian Inference to estimate initial cloud conditions, including temperature, final density, cosmic ray ionisation rate, and UV background radiation.
 - * Investigating the effect of incorporating survival function, CDF, and PDF for molecules with only upper-limit abundance measurements, improving uncertainty estimation of molecules with weak observation.
- Studying the Role of Galaxy Spheroids in Star Formation** *May 2025 – Ongoing*
 - * Supervised by Dr. Dario Colombo and Prof. Frank Bigiel at the Argelander-Institut für Astronomie (AIfA).
 - * Built a Python pipeline for multicomponent photometric decomposition of 643 galaxies (**iEDGE** survey), extracting Bulge-to-Total (B/T) light ratios and photometric bulge masses.
 - * Testing the influence of bulge mass on star formation quenching, thus testing the significance of morphological quenching.
- Scattering of Methanol Masers and Electron Distribution of the Milky Way** *December 2024 – April 2025*
 - * Research intern under Dr. Andreas Brunthaler and Prof. Amélie Saintonge at Max Planck Institute for Radio Astronomy, Bonn. Reduced and analysed **VLBA** data using **AIPS** and **ParselTongue** to measure angular broadening of 21 CH₃OH masers and 10 background quasars.
 - * Compared measured scattering against the **NE2001** electron density model via dispersion measure calculations. Observed discrepancies between the model predictions and the measured scattering for masers, which are attributed to intrinsic maser sizes. No discrepancies were observed for quasars.
- Detecting and Classifying AGNs in MaNGA Galaxies using Resolved BPT Diagnostics** *July 2024 - September 2024*
 - * Developed a **Python+SQL** pipeline to detect AGNs from the **MaNGA** survey by making resolved BPT maps and classifying AGNs based on H α equivalent width into strong/weak.
 - * Detected 567 AGNs from 10,070 MaNGA galaxies, with good classification overlap with broad, X-ray, and IR AGN catalogues; this project was supervised by Dr. Mallory Thorp and Prof. Frank Bigiel at the Argelander-Institut für Astronomie (AIfA).

Teaching Experience

- Argelander-Institut für Astronomie (AIfA)** August 2025 - November 2026
Wissenschaftliche Hilfskraft (Tutor of Optical Astronomy Exoplanet Transit Detection Lab), Bonn, Germany
 - * Explaining the observation, calibration, and data reduction of a modern optical telescope, conducting oral examinations, carrying out telescope observation, supervising data reduction, and grading lab reports of MSc students. Additionally, giving tours of the telescope on the rooftop of AIfA to visiting students and guests.
 - * Contributing to the astrophotography done at AIfA by observing Milky Way objects and other deep-sky objects.
- Naxxatra Sciences and Collaborative Research** January 2023 - February 2023
Research and Teaching Fellow, Bangalore, India
 - * Conducted hands-on astrophysics lectures for 300 students (grades 5-12), on topics like galaxy morphology, the cosmic distance ladder, the working of different telescopes, methods to detect exoplanets, and introduced dark matter and CMB.
 - * Performed experiments like sunspot tracking, acceleration due to gravity measurement, and Earth's circumference calculation. Supervised Solar observations. Conducted stargazing sessions and comet C/2022 E3 (ZTF) observation nights for students, faculty, and parents.

Scientific Training

- Attended the 2nd European ALMA Winter School 2026** January 2026
The school was organised by the University of Leiden, which covered the calibration, imaging, proposal writing, and reduction pipeline to study the astronomical data observed with the ALMA telescope. I was one of the few master's students to attend this school. I gained an improved understanding of imaging beyond the transitional CLEAN method.
- Attended the Physics of the Early Universe 2022** January 2022
I was one of the very few undergraduate students selected to attend this program organised by the International Centre for Theoretical Sciences (ICTS) Bangalore. Studied advanced topics such as Perturbation Theory, Dark Energy, Inflation, CMB anisotropies, Galaxy Formation, and Big Bang Nucleosynthesis.

Honours and Awards

- Selected for the International Summer School on the ISM of Nearby Galaxies** July 2025
I was selected to attend the International Summer School on the ISM of Nearby Galaxies 2025 with partial funding. Unfortunately, I could not attend the school as one of my MSc exams clashed with the school dates.
- Discovered Two Asteroids** September 2021
I detected near-Earth objects and main-belt asteroids in data from the Pan-STARRS survey, provided by the International Astronomical Search Collaboration (IASC). Made 12 preliminary discoveries and 2 provisional discoveries, including two main belt asteroids, namely, **2021 RZ8** and **2021 QO24**.

Skills

- * **Programming Languages:** Python, SQL, C++, HTML, Fortran
- * **Python Libraries:** NumPy, Pandas, Astropy, Matplotlib, Seaborn, SciPy, pymysql, ParselTongue, JAX, NumPyro, PyTorch, Scikit-learn, sep, photutils, astroquery
- * **Astrophysics Software:** CASA, AIPS, MESA, DS9, PHOEBE, Period04, Astrometrica, NINA
- * **Surveys & Telescopes:** MaNGA, BeSSeL, iEDGE, SDSS, Pan-STARRS, VLBA, ALMA
- * **Other Computational Skills:** Linux, L^AT_EX, GitHub, Microsoft Office
- * **Soft Skills:** Pedagogy and Science Communication, Leadership and Team Collaboration, Analytical Problem Solving, Rapid Skill Acquisition, Resilience and Discipline, Detail-Oriented and Self-Motivated

* **Languages:** Hindi (Native Speaker), English, German, Punjabi, and Marathi.

Other Relevant Experience

1. **Astrophotography at Universität Bonn**

August 2025 - Ongoing

Link: Astrophotography Jatin Chopra

Conducted frequent deep sky observations using the 35 cm optical telescope at AIfA. Observed objects like NGC 891, M51, M81, M82, M1, and NGC 7331.

2. **Writer at The Universe 101**

December 2020 - April 2021

Website: The Universe 101

Began an astrophysics blog to explain cosmic mysteries in simple terms. Articles like "Cosmic Alchemy" and "Dark Matter" received decent attention. I plan to restart the blog and eventually transform it into a YouTube channel.

References

* **Prof. Frank Bigiel**

Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
Professor of Astrophysics | bigiel@astro.uni-bonn.de

* **Prof. Serena Viti**

Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
Professor of Astrochemistry | sviti@uni-bonn.de

* **Dr. Mallory Thorp**

Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
Postdoctoral Researcher | mthorp@astro.uni-bonn.de

* **Dr. Kiran Wani**

Indian Institute of Astrophysics, Bangalore
Postdoctoral Researcher | kiran.wani@iiap.res.in